

J-space Phase 3 — mechanism, generalization, paper hardening working handout

branch `interp-jspace_part2` · run `root phase3_20260729` · 2026-07-30, VM10 late block

Tier legend. Every number carries its evidence tier: `PHASE2-CONFIRMATORY` results are frozen and immutable; `PHASE3-DEVELOPMENT` results guide design and are NOT confirmatory claims; `METHODS` results validate instruments. Nothing here exists outside `reports/evidence_events.jsonl`.

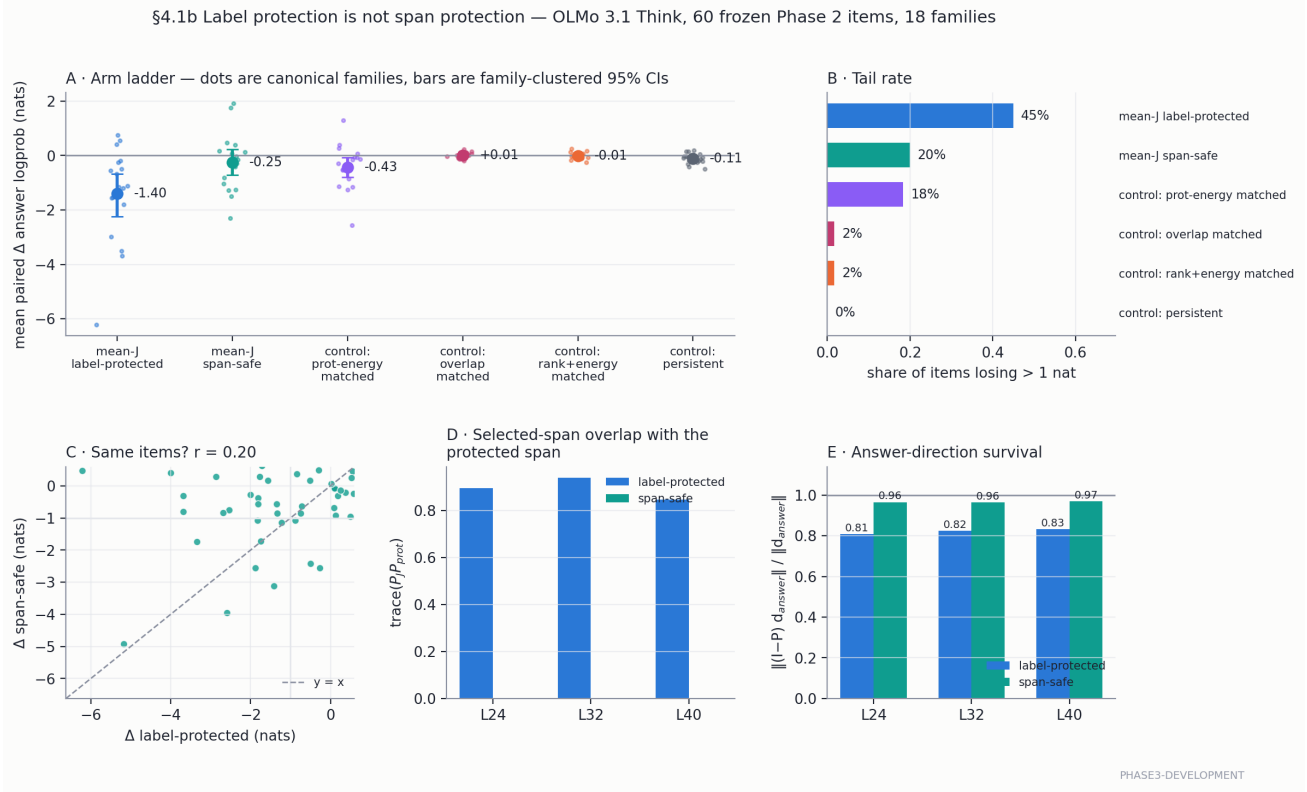
1 Where Phase 3 stands

Phase 2 closed with both preregistered primaries rejecting (HP1 task-shape contrast $-0.504 [-0.720, -0.295]$; HP3 Qwen J-specific tail $+0.279 [+0.205, +0.361]$, replicating held-out at $+0.297$) `PHASE2-CONFIRMATORY`. Phase 3 attacks the four gaps that block the strong manuscript sentence: the thin two-hop leg, the label-vs-span protection asymmetry, the missing prose exact control, and full clean-room reproduction.

- **§4.1a overlap mining** `p3-overlap-mining-v1` — done.
- **N8 Level 1 clean room** `p3-n8-level1-repro-v1` — done, all four primaries reproduce to $\leq 5e-4$.
- **§4.1b span audits** — done on ALL THREE primaries + cross-model synthesis `p3-span-audit-cross-model-v1`.
- **Workstream C prose exact control** — battery + machinery banked; model grids running now.
- **Banks F/S** — 54 of 72 target families authored, validated, and source-verified.
- Ahead: remaining bank tranches → `family_split_v2` → G5 → power sim → **Phase 3 freeze** → primary grid.

2 §4.1a — Selection pressure is a model property, not a damage predictor `phase3-development`

Protected-row selection pressure (clean top- k rows the J selection would have taken) does not predict per-item damage in any model (ρ 0.00–0.11, CIs cross 0), *but* it is a 12× model property: median blocked rows/position Think 1.33, Instruct 1.22, Qwen **0.10**. Accessibility signs flip between clusters: $\rho(\text{clean rank}, \Delta)$ $-0.30/ -0.31$ on the OLMo pair (CI-clean) vs $+0.24$ on Qwen. `p3-overlap-mining-v1`, fig. `p3f01`.



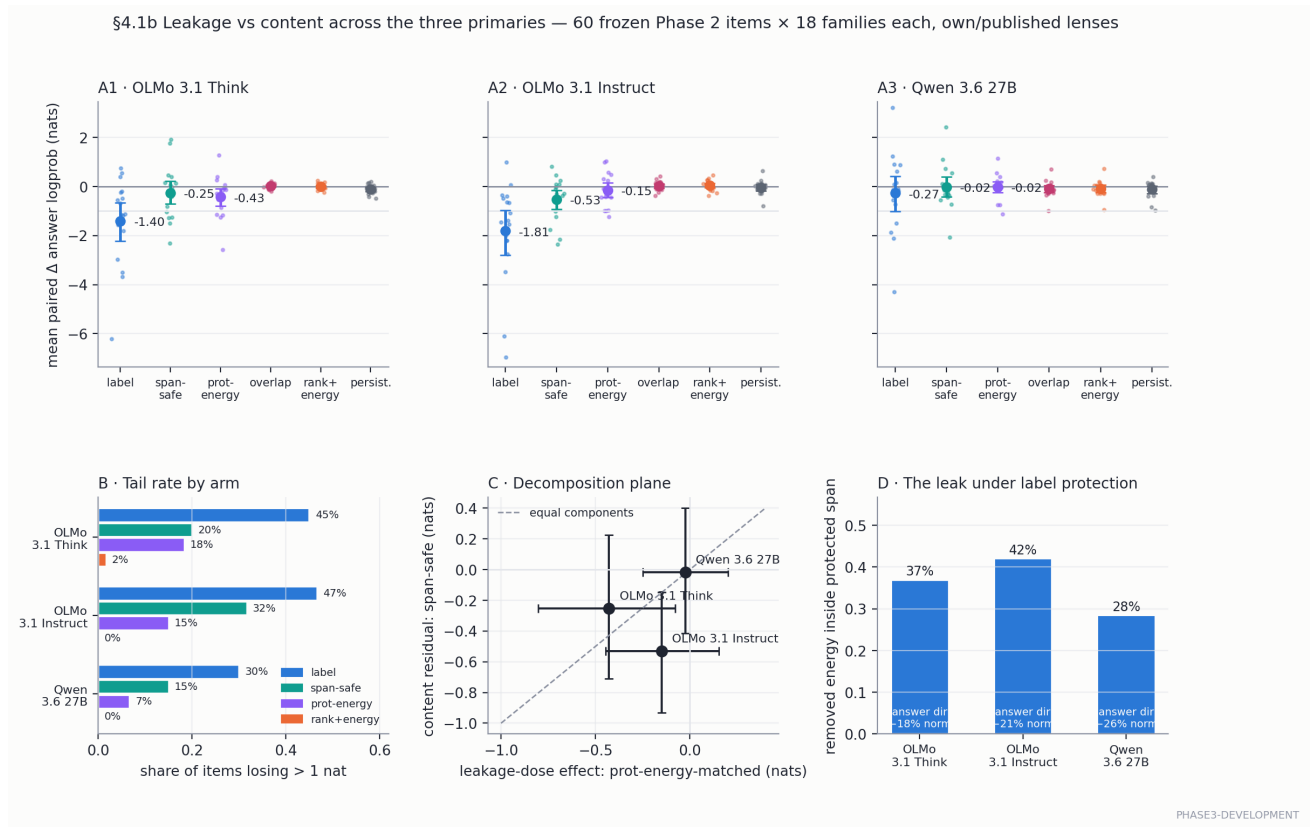
3 §4.1b — Label protection is not span protection

phase3-development

The paper-faithful label-protected arm *leaks into the very span it declares protected*, on every model (fig. above: OLMo 3.1 Think own lens, 60 frozen Phase 2 items). Span-safe protection zeroes the overlap exactly (its own positive control) and the effect decomposes:

family-weighted Δ (nats) · tail@-1	Think	Instruct	Qwen
label-protected	-1.40 [-2.24, -0.67] · 45%	-1.81 [-2.81, -0.97] · 47%	-0.27 [-1.01, +0.42] · 30%
span-safe	-0.25 [-0.71, +0.23] · 20%	-0.53 [-0.93, -0.15] · 32%	-0.02 [-0.41, +0.40] · 15%
prot-energy-matched	-0.43 [-0.80, -0.08] · 18%	-0.15 [-0.45, +0.16] · 15%	-0.02 [-0.25, +0.20] · 7%
span-safe own control	-0.005 · 2%	+0.01 · 3%	-0.08 · 2%
specific span-safe tail	+18pp	+28pp	+13pp
r (label, span-safe) per item	0.20	0.29	0.75
removed energy inside protected span	37%	42%	28%
answer-direction norm lost	18%	21%	26%

Reading. (1) The leak *geometry* is universal — every model deletes protected-span energy under label protection. (2) The *behavioral* leakage component is essentially a Think phenomenon (prot-energy control reproduces $\sim 1/3$ of its label mean, nil on Qwen). (3) The content residual is largest and CI-clean on Instruct. (4) A J-specific tail survives full span protection on all three (own controls $\leq 3\%$). (5) On Qwen, span-safe *rescales* the same items ($r = 0.75$); on OLMo it *reallocates* — so the decomposition must be carried as two components, never one number. Phase 2’s HP3 transfers nearly intact on Qwen, as §4.1a’s selection-pressure contrast predicted. p3-span-audit-{olmo31-think-v2,olmo31-instruct-v1,qwen36-27b-v1} fig. p3f03.



⇒ `meanJ_span_safe` is the P3-P2 primary arm; `prot_energy_matched` its preregistered comparator. Scope: development tier; the sharp control was designed after seeing the Think overlap numbers (fine for arm selection, disqualifying for confirmatory claims); confirmatory versions run on the thick bank post-freeze.

4 Instrument notes

methods

- **N8 Level 1 passes:** a narrative-blind agent regenerated the full Phase 2 primary table from frozen parquets in a clean output root; all four quantities match to $\leq 5e-4$. Levels 2–3 (sentinel subsets, one full GPU cell) remain open as release gates. `p3-n8-level1-repro-v1`
- **BOS units on Qwen:** `ScoringSession` demanded a `bos_token_id`; Qwen has none, and `jlenz.from_hf(force_bos=True)` is a no-op there — every frozen Phase 2 Qwen artifact is in *native* units. The session now scores native exactly when no BOS exists and records the applied convention (commit 70d8eae).
- **Units verified bit-exactly:** all three span audits reproduce frozen N6 per-alias baselines exactly (60/60 items per model). Where an audit baseline differs from N6’s `lp_canonical`, the audit’s alias choice named a different case variant — same machinery, same units.
- **Report CI correction:** the Think §4.1b section’s first published label/span-safe/prot-energy CIs matched no registered artifact (transcribed from a mid-audit dev render). `REPORT_PHASE3` now quotes the registered `p3-span-audit-figure-olmo31-think-v2` values (as does the table above); point estimates, tails, and conclusions unchanged.

5 Workstream C — the exact control on prose

phase3-development

Phase 2’s prose guard skipped the primary matched control (it consumes the J arm’s per-position profile); §7.1 makes it mandatory before any “selective” wording about Instruct (PI resolution R4). Now built and running:

- **Guard battery v2** `p3-guard-battery-v2`: 104 items / 8 domains (wikitext-test prose + long-context, wikipedia leads, UltraChat dialogue, the-stack code + SQL + markdown docs, 20 in-repo grammar minimal pairs), source-pinned; wikitext TEST held out from every lens-fit corpus.
- **Grid:** clean → label-J (profile logged) → span-safe-J → exact instant rank+energy matched (consumes profile) → prot-energy-matched → mechanics random → logit-protected; NLL/token, KL-from-clean, top-1 agreement,

entropy, EOS; grammar preference margins; pre-registered greedy generation audit.

- Think grid is running at the time of writing; Instruct and Qwen follow in this window. Figure p3f04 + cross-model statistics land when all three bank.

6 Banks F and S — 54 of 72 families

phase3-development

- **Bank F v5** `p3-bank-f-v5`: **164 bundles / 38 families**, every bundle direct/composed/bridge-supplied + rotated counterfactual, functional first hops with per-family ambiguity notes, cross-phase dedup against every Phase 2 (bridge, answer) pair AND bare answer (858 burned answers), one (bridge, answer) pair per bank, distinct templates per family. **164/164 verified** against a pinned reference.
- **Bank S v2** `p3-bank-s-v2`: **80 bundles / 16 synthetic template families** (lookup, binding, path, distractor, reversible-mapping types per §5.2) — the working-memory bank whose composed—direct contrast is preregistered as Family B’s first member (addendum §4.3).
- **Verification pivot** `p3-wiki-reference-v4`: the Drive `wikimedia/wikipedia 20231101.en` snapshot is hub-authentic (shard sha256 verified) yet *missing the plain articles of major entities* — ‘Thailand’ is absent from all 6,407,814 rows while “Thailand at the 1992 Summer Olympics” exists; the release evidently drops its largest pages. Bank verification therefore runs against freshly fetched wikitext pinned by *per-page revision ids* (strictly stronger provenance than a dump date). 248/248 pages fetched; every hop of every shipped bundle is stated on its source page.
- Remaining to the §5.3 floor: ~18 families (tranche 4) + top-ups to ≥ 5 bundles/family before G5 pruning; then `family_split_v2` (seed-active), G5 scoring $\times 3$ models (scorer `g5_bank_scoring.py` is written), power sim, freeze.

7 Paper tree

Sections 1–3 (problem, assay repair, Phase 2 matrix) are drafted from frozen material only, with evidence ids inline and the §0.2 calibrated sentence verbatim (`paper/01--03*.md`). Writing them surfaced no overclaims to withdraw; the assay-repair section doubles as the methods-contribution narrative.

Standing rules: Phase 2 frozen and immutable · producers refuse dirty trees · supersede, never edit · tier watermarks everywhere · the contradiction heuristic (it caught the BOS guard and the wikipedia-release defect this block) · prohibited claims per nextsteps §4.5.